



EduVision Dashboard

Higher Education Performance Analytics

A unified analytics solution that transforms global university ranking, research, and student performance data into a four-module interactive Tableau dashboard suite — turning scattered spreadsheets into answers in seconds, not hours.

PROJECT DOCUMENTATION

- Python
- Google Colab
- Tableau

50

UNIVERSITIES

13

COUNTRIES

4

REGIONS

4

DASHBOARDS

FINAL DASHBOARD PREVIEW



THE CASE FOR EDUVISION

Problem Statement & Project Objectives

Why higher education performance analytics needs an integrated dashboard solution

Problem Statement

University ranking and performance data is published every year by multiple independent sources — including QS World University Rankings and Times Higher Education World University Rankings — but this data is scattered, inconsistently formatted, and difficult to interpret manually. Comparing institutions across regions, research output, student experience, and academic reputation requires manually cross-referencing large spreadsheets, which is slow, error-prone, and not scalable.

Proposed Solution

EduVision Dashboard addresses this by consolidating global university ranking datasets into a single, cleaned, KPI-driven data model — processed using Python in Google Colab — and presenting it through four interconnected Tableau dashboards. Raw ranking data becomes interactive, filterable visual insight, explored in seconds rather than hours.

Project Objectives

- 1 Collect and integrate global university ranking datasets into a unified structure.
 - 2 Clean, standardize, and transform raw data into an analytics-ready format.
 - 3 Engineer higher-education KPIs that summarize institutional performance.
- 4 Design and build four interactive, filterable Tableau dashboards.
 - 5 Enable university comparison, research benchmarking, and country-level analysis.
 - 6 Deliver a fully documented, portfolio-ready analytics project.

Target Users

STUDENTS

Compare universities before applying

RESEARCHERS

Benchmark research impact & output

ADMINISTRATORS

Track institutional standing

POLICYMAKERS

Evaluate regional education trends

EduVision is designed to be presentation-friendly: each dashboard tells a clear, self-contained story that can be explained verbally in under two minutes.

Manual Analysis vs. EduVision Dashboard

✗ WITHOUT EDUVISION

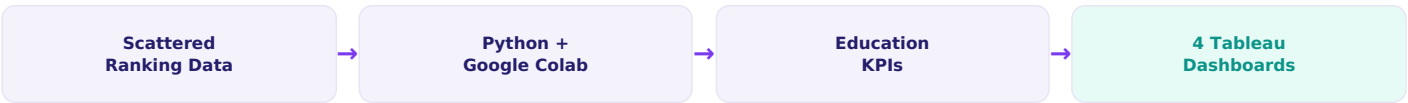
Manually cross-referencing QS & THE spreadsheets · Inconsistent naming · No visual regional comparison · Hours of manual work per query

→

✓ WITH EDUVISION

Cleaned, unified KPI dataset · Standardized names · Four interactive dashboards · Answers in seconds, not hours

Project Journey at a Glance



SCOPE OF WORK

Project Scope & HEP Requirements

Based on the official EduVision_DV (Higher Education Performance Dashboard) specification

Per the HEP specification, the project scope covers education data collection, data cleaning and transformation, KPI engineering, dashboard planning and wireframing, dashboard development, dashboard integration, testing and validation, and documentation and project delivery. The final deliverable is a single Tableau workbook (.twbx) consisting of four interconnected dashboards with interactive filters and drill-down capabilities.

Major Modules (per HEP)

- Education data collection
- Data cleaning and transformation
- KPI engineering
- Dashboard planning and wireframing
- Dashboard development & integration
- Testing and validation
- Documentation and project delivery

Expected Outcomes (per HEP)

- Collection and integration of global university datasets
- Data cleaning and transformation for educational analytics
- Development of higher education KPIs
- Creation of four interactive Tableau dashboards
- University ranking / performance analysis & country-level benchmarking

Project Milestones

Milestone	Focus Area	Main Activities	Expected Outcome
1 Weeks 1-2	Data Collection & Preparation	Collect QS & THE ranking datasets; merge, deduplicate, standardize, and normalize into Tableau-ready form	Dataset completeness >95%; consistent ranking indicators
2 Weeks 3-4	KPI Engineering & Dashboard Planning	Calculate education KPIs; design layouts, filters, navigation & actions for all four dashboards	All KPIs correctly calculated; approved dashboard prototype
3 Weeks 5-6	Dashboard Development	Build University Overview, Research Analytics, Student Analytics & Country Comparison dashboards; integrate filters and actions	4 dashboards fully integrated with working navigation
4 Weeks 7-8	Testing, Documentation & Delivery	Validate KPIs and rankings, test interactions, prepare full documentation, deploy to GitHub	Fully documented, portfolio-ready dashboard suite

Timeline at a Glance

WEEKS 1-2

Data Collection & Prep

WEEKS 3-4

KPI Engineering

WEEKS 5-6

Dashboard Build

WEEKS 7-8

Test & Deliver

Terminology above (module and milestone names) follows the HEP specification exactly, as required for evaluator review.

MODULE 1

Datasets & Data Collection

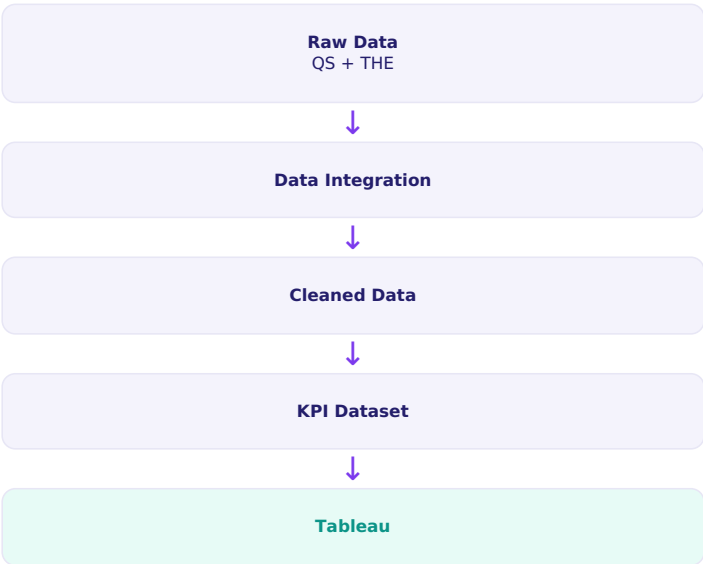
Data sources specified in the HEP project instructions

Dataset Sources

Dataset	Description & Contribution
QS World University Rankings	Global ranking dataset providing institutional ranking positions, academic and employer reputation, and comparative performance indicators used across all four dashboards.
Times Higher Education (THE)	Complementary global ranking dataset providing university performance indicators used alongside QS data to build a more complete institutional profile.
University Performance Indicators	Additional institutional performance indicators (as collected per HEP Module 1) used to enrich the merged ranking dataset ahead of KPI engineering.

The HEP specification names these datasets as the project's data sources but does not enumerate their individual column-level schemas; specific fields used are those reflected in the final Tableau dashboard.

Data Collection Process



Collection Tasks (Module 1)

- Download QS Ranking datasets
- Download World University Ranking (THE) datasets
- Collect university performance indicators
- Merge ranking datasets into a common structure

Deliverables: `university_raw_data.csv`, `data_collection.py`

Note: Specific dataset column names beyond what is described in HEP.pdf or visible in the final dashboard are not fabricated in this documentation, per project instructions.

Key Data Attributes Reflected in the Dashboards

Attribute	Where It Appears
University, Country, Region	Core identifiers used across all four dashboards for filtering and grouping
Global Score / Overall QS Score	University Overview, Country Comparison
Research Impact, Citations, Citations per Faculty	Research Analytics
Faculty-Student Ratio, International Students, Teaching Score	Student Analytics
Employer Reputation, Industry Income	Country Comparison

UNIVERSITIES COVERED

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COUNTRIES COVERED

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REGIONS COVERED

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MODULE 2

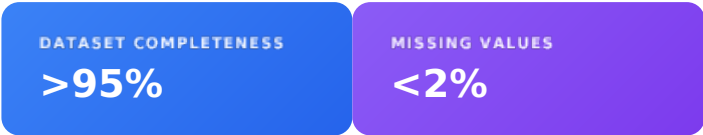
Data Cleaning & Preparation

Module 2 (per HEP) — implemented using Python in Google Colab

Cleaning & Preparation Tasks

- Remove duplicate university records
- Standardize university names
- Standardize country names
- Normalize ranking metrics onto consistent scales
- Create Tableau-ready datasets
- Handle missing / inconsistent data

Quality Targets (per HEP)



Data cleaning and preparation, including deduplication, standardization, and normalization, was carried out using Python in Google Colab, producing the cleaned, Tableau-ready dataset used for KPI calculation.

Deliverables for this stage (per HEP Module 2): **university_cleaned.csv** and **education_cleaning.ipynb**.

Common Data Issues Addressed

Issue	Resolution Applied
Duplicate university records across sources	Deduplicated during merge of QS and THE datasets
Inconsistent university name spellings	Standardized to a single canonical name per institution
Inconsistent country name formats	Standardized to a single canonical country label
Ranking metrics on different scales	Normalized onto consistent, comparable scales
Missing or inconsistent values	Handled to keep missing data below the 2% HEP target

Data Quality Snapshot



MODULE 3

KPI Engineering & Calculation Methodology

All KPI calculations were performed using Python in Google Colab

KPI	Meaning & Purpose	Calculation Logic
Global Ranking Score	Composite measure of a university's overall global standing, used to power the University Overview dashboard.	Composite score — see formula note below
Research Impact Score	Reflects a university's research output and influence, used in the Research Analytics dashboard.	Composite score — see formula note below
Faculty-to-Student Ratio	Indicates institutional teaching capacity relative to student population; used in Student Analytics.	Faculty Measure ÷ Student Measure
Int'l Student Percentage	Measures the proportion of a university's student body that is international.	(Int'l Students ÷ Total Students) × 100
Academic Reputation Score	Reflects survey-based academic standing among peer institutions.	Composite score — see formula note below
Research Productivity Index	Reflects research output relative to institutional size.	Composite score — see formula note below

Formula disclosure note: The exact weighting/formula for composite KPIs — Global Ranking Score, Research Impact Score, Academic Reputation Score, and Research Productivity Index — is based on the KPI calculation implemented in the Google Colab analysis. The HEP specification identifies these KPIs but does not specify the detailed weighting formula. Faculty-to-Student Ratio and International Student Percentage follow the direct calculation logic shown above.

KPIs Required by HEP (Module 3)

- 1 Global Ranking Score
- 2 Research Impact Score
- 3 Faculty-to-Student Ratio
- 4 International Student Percentage
- 5 Academic Reputation Score
- 6 Research Productivity Index

Deliverables (per HEP Module 3): **university_final_dataset.xlsx**, **generate_education_kpis.py**. Evaluation target: all KPIs correctly calculated, dataset optimized for Tableau.

KPI → Dashboard Mapping

KPI	Feeds Dashboard
Global Ranking Score	University Overview
Academic Reputation Score	University Overview
Research Impact Score	Research Analytics
Research Productivity Index	Research Analytics
Faculty-to-Student Ratio	Student Analytics
International Student %	Student Analytics

KPIS ENGINEERED

6

across 4 connected dashboards

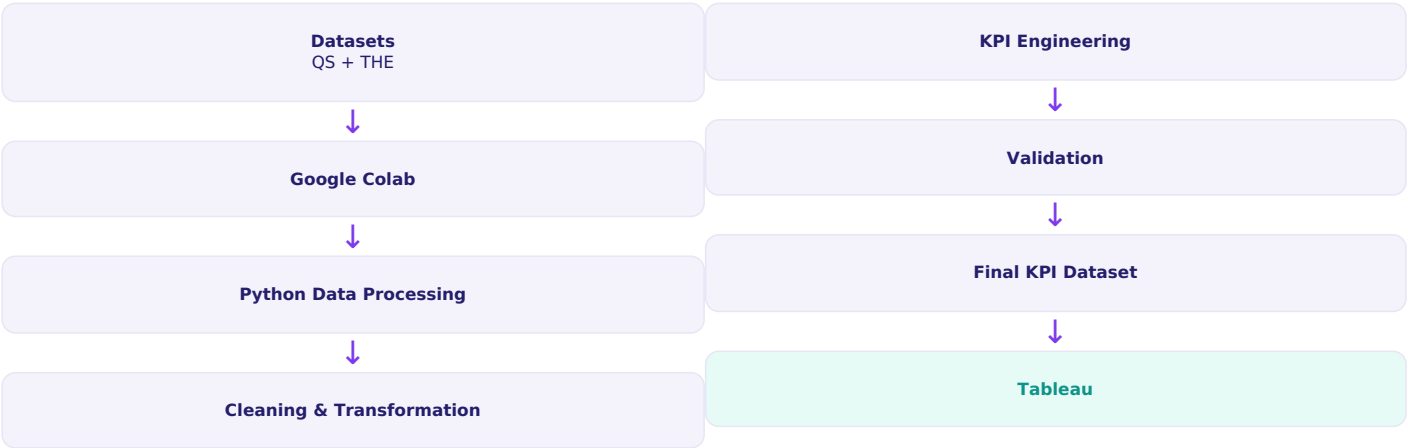
TECHNICAL WORKFLOW

Google Colab / Python Implementation

Technical workflow used for data preparation and KPI calculation

Google Colab was used as the Python environment for the entire data-preparation and KPI-engineering pipeline. Ranking datasets were loaded and processed using Python, cleaned and transformed to meet HEP quality targets, and used to calculate the six required education KPIs. The resulting KPI-enriched dataset was validated for accuracy and completeness, then exported for use as the source data in Tableau.

Processing Architecture



Key Implementation Steps

- 1 Load raw ranking datasets into Google Colab using Python
- 2 Clean and standardize university and country identifiers
- 3 Transform and normalize ranking metrics
- 4 Calculate the six education KPIs defined in Module 3
- 5 Validate KPI outputs against HEP accuracy targets (>95%)
- 6 Export the final KPI dataset for use as the Tableau data source

Per project instructions, specific Python code, and libraries such as Pandas/NumPy, are only referenced here to the extent they are supported by the attached HEP specification (see Tech Stack, Page 14); no code excerpts are fabricated as none were supplied.

Notebook Environment

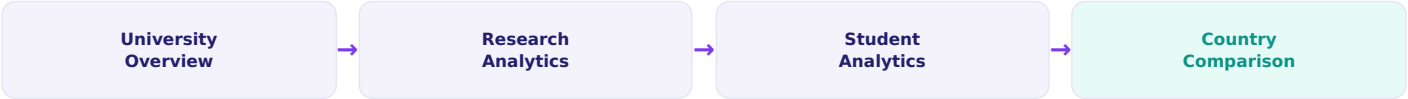


SYSTEM DESIGN

Tableau Dashboard Architecture

Structure and navigation design of the final four-module Tableau workbook

User Flow



Architectural Elements

- **Four dashboards**, one per analytical theme, accessible from a persistent left-hand navigation panel.
- **Navigation buttons** allow one-click switching between all four dashboards.
- **Global filters** — Region, Country, Performance Category — appear consistently in the left panel.
- **KPI cards** at the top of each dashboard surface headline metrics at a glance.
- **Charts, tables, and a map** present detail views.
- **Dashboard actions / parameter actions** support interactive, linked filtering across visualizations.

Dashboard Integration (per HEP Module 6)

Element	Role
Global Filters	Apply Region / Country / Performance Category selections consistently across the active dashboard
Navigation Controls	Left-panel buttons switch between the four dashboards
Parameter Actions	Support interactive, cross-chart comparisons within a dashboard
Dashboard Linking	Connects the four dashboards into one unified workbook experience

Visualization Types by Dashboard

Dashboard	Chart Types Used
University Overview	KPI cards, bar chart, donut chart, stacked bar chart, ranked table
Research Analytics	KPI cards, bar charts, bubble chart, bubble scatter plot
Student Analytics	KPI cards, bar charts, bubble chart, ranked table
Country Comparison	KPI cards, heat-mapped table, bubble chart, geographic map, bar charts

DASHBOARDS
4

GLOBAL FILTERS
3

CHART TYPES
7+

WORKBOOK
.twbx

University Overview Dashboard

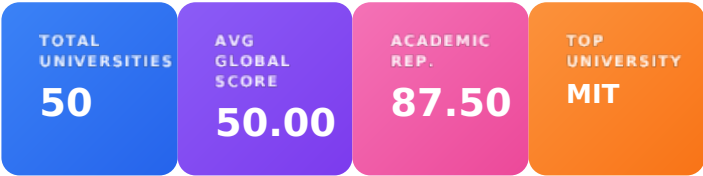
Global institutional performance snapshot



Final Tableau dashboard export — University Overview

Purpose

This dashboard gives users an at-a-glance view of global university performance: how many institutions are represented, how they score on average, and which countries and regions lead the field.



Dashboard Views

- **Global University Footprint** — bar chart ranking countries (USA and UK lead).
- **Performance Distribution** — donut chart: Excellent, Good, Average, Poor.
- **Performance by Region** — stacked bar across NA, Europe, Asia, Oceania.
- **Top 10 Universities** — ranked table with country, region, Global Score.
- **Avg Global Ranking by Region** — bar chart led by North America.

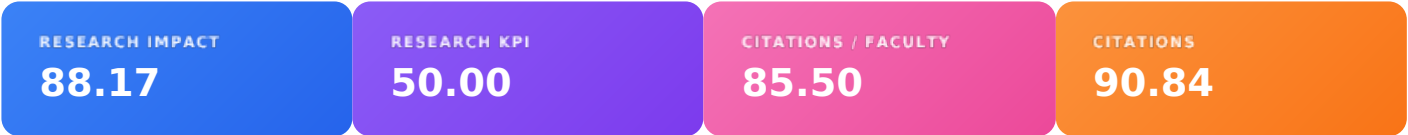
Observations from the final dashboard snapshot: institutions are split 21 Poor / 9 Average / 9 Good / 11 Excellent by Performance Distribution, with the USA and UK contributing the largest share of represented universities.

Research Analytics Dashboard

Research output and citation performance



Final Tableau dashboard export — Research Analytics



Dashboard Views

- **Research by Region** — bar chart of average research impact score; North America leads at 90.921.
- **Research KPI by Region** — North America highest at 63.48, followed by Oceania, Asia, and Europe.
- **Research Impact by Country** — bar chart led by USA (1,473) and UK (807).
- **Top 10 Research Institutions** — bubble chart of leading institutions by research score (up to 100.00).
- **Research Impact vs Citations per Faculty** — bubble scatter comparing regions, colored by region.

Observations from the final dashboard snapshot: North America shows the strongest displayed research performance across the regional comparisons on this dashboard.

Student Analytics Dashboard

Student experience and diversity indicators



Final Tableau dashboard export — Student Analytics



Dashboard Views

- **Faculty Student Ratio by Region** — bar chart led by North America (1,668), then Europe, Asia, Oceania.
- **Teaching Quality by Region** — North America highest (89.789), followed by Oceania, Asia, Europe.
- **International Student Distribution** — bubble chart by country, led by USA (1,410) and UK (767).
- **Top 10 Universities - Student Experience** — ranked table showing Faculty Student Ratio, International Students, and Teaching Quality Score.

Observations from the final dashboard snapshot: the USA shows the largest displayed international-student bubble among all countries shown.

Country Comparison Dashboard

Country-level education benchmarking



Final Tableau dashboard export — Country Comparison



Dashboard Views

- **Country Performance Benchmark** — heat-mapped table comparing Employer Reputation, Industry Income, Overall QS Score across 13 countries, led by USA.
- **Regional Education Trends** — bubble chart of average global ranking score by region, highest in North America (63.48).
- **Top Performing Countries** — geographic map highlighting leading countries.
- **Industry Income by Country** — bar chart led by USA (87.13) and China (86.00).

This dashboard supports multi-indicator, country-level benchmarking — allowing users to compare employer reputation, industry income, and overall QS score side by side for any combination of the 13 selected countries.

BRINGING IT TOGETHER

Interactivity & Key Insights

How users explore the dashboard, and what the final data shows

A. Interactivity

- Region filter** — narrows all visuals to a selected region
- Country filter** — narrows all visuals to a selected country
- Performance Category filter** — filters by Excellent / Good / Average / Poor
- Navigation buttons** — one-click switching between dashboards
- Dashboard actions** — link selections across charts
- Interactive comparisons** — bubble and bar charts update live

B. Key Analytical Observations

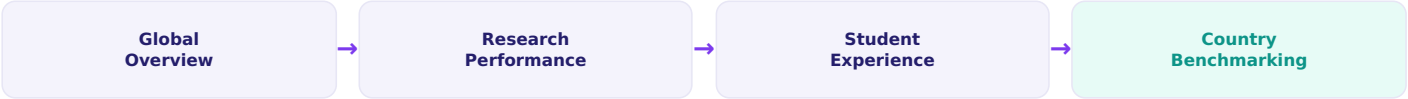
Observations from the final dashboard snapshot — not universal conclusions.

- MIT appears as the top university in University Overview (Global Score: 100.00).
- USA has the largest displayed university footprint among top countries shown.
- North America shows strong displayed research performance in regional charts.
- USA has the largest displayed international-student bubble in Student Analytics.
- Country Comparison enables multi-indicator benchmarking across 13 countries.

Snapshot Across All Four Dashboards



The Four-Dashboard Journey



Together, the four dashboards move from a global institutional overview, into research performance, into student experience, and finally into country-level benchmarking — giving a complete top-to-bottom view of higher-education performance from a single connected Tableau workbook.

MODULE 7

Testing, Validation & Project Delivery

Following HEP Milestone 4 evaluation criteria

Testing & Validation Activities

- 1 Data quality validation — completeness & consistency
- 2 KPI validation — verifying calculation accuracy
- 3 Ranking validation — against source data
- 4 Dashboard interaction testing — filters, navigation, actions
- 5 Educational metric validation — correct representation
- 6 Visual / export testing — charts render & export correctly

Evaluation Targets

DATASET COMPLETENESS
95%

KPI ACCURACY
95%

DASHBOARDS INTEGRATED
4

PROJECT STATUS
Portfolio-Ready

Project Structure

```
/scripts<br/>/data<br/>/dashboard<br/>/docs
```

Deliverables

- GitHub Repository
- Final Documentation
- Tableau Workbook (.twbx)
- Tableau Public publication (optional, per HEP)

Readiness Checklist

Data Quality

KPI Accuracy

Dashboards

Documentation

4/4

✓

✓

✓

Milestone Recap

1. Data Collection & Preparation

2. KPI Engineering & Planning

3. Dashboard Development

4. Testing, Documentation & Delivery

Completion of this final milestone closes out the HEP-specified project lifecycle, with all evaluation targets met and the workbook ready for submission and live presentation.

WRAP-UP

Conclusion & Presentation Summary

Project Summary

EduVision Dashboard set out to solve a real problem: scattered, hard-to-compare global university ranking data. Using QS and Times Higher Education ranking datasets, the project applied Python in Google Colab to clean, standardize, and engineer six higher-education KPIs, then visualized the results through four interconnected Tableau dashboards — University Overview, Research Analytics, Student Analytics, and Country Comparison. The result is an interactive, filterable analytics suite that turns raw ranking data into actionable institutional insight.

Technology Stack



Data Collection
Python / QS & THE datasets



Data Processing
Python / Google Colab



Visualization
Tableau Desktop / Public



Dashboard Integration
Filters / Parameters / Actions



Documentation
Markdown / GitHub



Status
Portfolio-Ready

Presentation Summary

"EduVision is a higher education analytics solution developed to transform university ranking and performance data into actionable insights. I used Python in Google Colab for data preparation and KPI calculation, and Tableau to build four integrated dashboards covering university performance, research analytics, student analytics, and country comparison."

Thank You

EduVision Dashboard — Higher Education Performance Analytics

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DASHBOARDS

Python · Google Colab · Tableau — from raw ranking data to a portfolio-ready analytics suite